LEON NSAMBA GRADIENT DESCENT

$y = m_1x_1 + m_2x_2 + b$

$\hat{y} = mx + b$, $m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, $m_1 = -1$, $m_2 = 2$, $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$, $x_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $x_2 = \begin{bmatrix} 3 \\ 10 \end{bmatrix}$,

$y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

x	y	$\hat{y}$	$\hat{y} - y$
$\begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$	$\begin{bmatrix} 5 \\ 6 \end{bmatrix}$	$\begin{bmatrix} 6 \\ 17 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 11 \end{bmatrix}$

Getting $\hat{y}$ (yhat)

$\hat{y} = mx + b$

$\hat{y} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\hat{y} = \left[\begin{pmatrix} (-1 \times 1) + (2 \times 3) \\ (-1 \times 4) + (2 \times 10) \end{pmatrix} \right] + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\hat{y} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$

$\hat{y} - y = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

$\hat{y} - y = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

Calculating the ~~loss~~ MSE

$MSE = \frac{1}{n} \sum_{i=1}^{n} (\hat{y} - y)^2$, $n = 2$

$MSE = \frac{1}{2} \begin{bmatrix} 1 \\ 11 \end{bmatrix}^2 \rightarrow \frac{1}{2} \left(\begin{bmatrix} 1 \\ 11 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 11 \end{bmatrix} \right)$

$MSE = \frac{1}{2} \left[(1 \times 1) + (11 \times 11) \right]$

$MSE = \frac{1}{2} \times 122$

$MSE = 61$

Calculating the loss function(s)

Let MSE be J

$\frac{dJ}{dm_1} = ?, \frac{dJ}{dm_2} = ?, \frac{dJ}{db} = ?$

From

$$J = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}-y)^2$$

$$y = m_1x_1 + m_2x_2 + b$$

$$J = \frac{1}{n}\sum_{i=1}^{n}(\hat{y} - m_1x_1 + m_2x_2 + b)^2$$

let $e = (\hat{y} - m_1x_1 + m_2x_2 + b)$

$$J = \frac{1}{n}\sum_{i=1}^{n} e^2$$

$\frac{dJ}{dm_1} = \frac{dJ}{de}\cdot\frac{de}{dm_1}$ → Applying the chain rule

$$\boxed{\frac{dJ}{de} = \frac{2}{n}\sum_{i=1}^{n} e} \quad -①$$

$e = \hat{y} - m_1x_1 + m_2x_2 + b$, m_1, m_2, x_2, b behave like constants hence equate to 0

$$\boxed{\frac{de}{dm_1} = 0 + x_1 + 0 + 0} \quad -②$$

$$\frac{dJ}{dm_1} = \frac{2}{n}\sum_{i=1}^{n} e \cdot +x_1$$

$$\frac{dJ}{dm_1} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - [m_1x_1 + m_2x_2 + b]) \cdot -x_1$$

→ y

$$\boxed{\frac{dJ}{dm_1} = +\frac{2}{n}\sum_{i=1}^{n}(\hat{y} - y)x_1} \quad -③$$

$\frac{dJ}{m_2} = ?$ (Getting $\frac{dJ}{dm_2}$)

$\frac{dJ}{dm_2} = \frac{dJ}{de} \cdot \frac{de}{dm_2}$

$\hat{y}-$
$e = x_1 m_1 + x_2 m_2 + b$

$\frac{de}{dm_2} = 0 + x_2 + b$

$\boxed{\frac{de}{dm_2} = x_2}$

$J = \frac{1}{n}\sum^{n} e^2$

$\boxed{\frac{dJ}{de} = \frac{2}{n}\sum_{i=1}^{n} e}$

Applying chain rule

for $\frac{dJ}{dm_2} = \frac{dJ}{de} \cdot \frac{de}{dm_2}$

$\frac{dJ}{dm_2} = \frac{2}{n}\sum_{i=1}^{n} e \cdot x_2$

where $\hat{y}-$
$e = x_1 m_1 + x_2 m_2 + b$

$\frac{dJ}{dm_2} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - \underbrace{(x_1 m_1 + x_2 m_2 + b)}_{\hat{y}}) \cdot x_2$

$\boxed{\frac{dJ}{dm_2} = \frac{2}{n}\sum^{n} (\hat{y} - y) x_2}$ — (11)

$\frac{dJ}{db} = ?$ (Getting $\frac{dJ}{db} = ?$)

$J = \frac{1}{n}\sum_{i=1}^{n} e^2$, where

$e = \hat{y} - y$

$e = \hat{y} - m_1x_1 + m_2x_2 + b$

$\frac{de}{db} = 0 - 0 + 0 + 1$

$$\boxed{\frac{de}{db} = 1}$$

$$\boxed{\frac{dJ}{de} = \frac{2}{n}\sum_{i=1}^{n} e}$$

Applying chain rule for $\frac{dJ}{db}$

$$\frac{dJ}{db} = \frac{dJ}{de} \cdot \frac{de}{db}$$

$$\frac{dJ}{db} = \frac{2}{n}\sum_{i=1}^{n} e \cdot 1$$

$$\frac{dJ}{db} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - \boxed{m_1x_1 + m_2x_2 + b}) \quad \leftarrow \hat{y}$$

$$\boxed{\frac{dJ}{db} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - y)} \quad \text{(11)}$$

$$\frac{dJ}{dm_1} = \frac{-2}{n}\sum_{i=1}^{n} (\hat{y} - y)x_1 \;,\; \frac{dJ}{dm_2} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - y)x_2 \;,\; \frac{dJ}{db} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - y)$$

$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ $m_1 = -1, m_2 = 2$

$\hat{y} - y = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$,

$x_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $x_2 = \begin{bmatrix} 3 \\ 10 \end{bmatrix}$

$n = 2$

$$m_1 new = m_1 old - \alpha \frac{dJ}{dm_1}$$

$$m_1 new = m_1 old - \alpha \left[\frac{+2}{n} \sum_{i=1}^{n} (\hat{y} - y) x_1 \right]$$

$$m_1 new = -1 - \alpha \left[\frac{+2}{2} \cancel{\sum_{i=1}^{n}} \begin{bmatrix} 1 \\ 11 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 4 \end{bmatrix} \right]$$

$$m_1 new = -1 - \alpha \left[+1 \; (1 \times 1) + (11 \times 4) \right]$$

$$m_1 new = -1 - \alpha \left[+45 \right]$$

$$\boxed{m_1 new = -1 \$ -45\alpha} \quad \text{—— (I)}$$

$$m_2 new = m_2 old - \alpha \left[\frac{2}{n} \sum_{i=1}^{n} (\hat{y} - y) x_2 \right]$$

$$m_2 new = 2 - \alpha \left[\frac{2}{2} \begin{bmatrix} 1 \\ 11 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 10 \end{bmatrix} \right]$$

$$m_2 new = 2 - \alpha \left[(1 \times 3) + (11 \times 10) \right]$$

$$m_2 new = 2 - \alpha \left[113 \right]$$

$$\boxed{m_2 new = 2 - 113\alpha} \quad \text{—— (II)}$$

$b = 1$

$$b_{new} = b_{old} - \alpha \frac{dJ}{db}$$

$$b_{new} = b_{old} - \alpha \left[\frac{3}{n} \sum_{i=1}^{n} (\hat{y} - y) \right]$$

$$b_{new} = b_{old} - \alpha \left[\frac{3}{2} \begin{bmatrix} 1 \\ 11 \end{bmatrix} \cdot 1 \right]$$

$$b_{new} = 1 - \alpha \left[(1 \times 1) + (11 \times 11) \right]$$

$$\boxed{b_{new} = 1 - 12\alpha} \quad (1)$$

$$\boxed{m_{1new} = -1 + 45\alpha, \; m_{2new} = 2 - 113\alpha, \; b_{new} = 1 - 12\alpha}$$

Let $\alpha = 0.01$

$m_{1new} = -1 + 45(0.01)$

$m_{1new} = -1 + 0.45$

$m_{1new} = -0.55$ ~~-1.45~~

$m_{2new} = 2 - 113\alpha$

$m_{2new} = 2 - 113(0.01)$

$m_{2new} = 0.87$

$b_{new} = 1 - 12\alpha$

$b_{new} = 1 - 12(0.01)$

$b_{new} = 0.88$

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (\hat{y} - y)^2$$

$\hat{y} = mx + b$

$$\hat{y} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} (-1.45 \times 1) + (-1.45 \times 3) \\ (0.87 \times 4) + (0.87 \times 10) \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} -5.8 \\ 12.18 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

MSE when $m_1 new = -1.45$, $m_2 new = 0.87$, $b new = 0.88$

$\hat{y} = mx + b$ $\quad m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$

$$\hat{y} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} (-1.45 \times 1) + (0.87 \times 3) \\ (-1.45 \times 4) + (0.87 \times (0) \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 1.16 \\ 2.9 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix} \qquad \hat{y} - y = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (\hat{y} - y)^2$$

$$MSE = \frac{1}{2} \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}^2$$

$$MSE = \frac{1}{2} \left[(-2.96)^2 + (-2.22)^2 \right]$$

$MSE = 6.85$